

We list all these risks in Table 2 and analyse them for occurrence, severity, and detection as we have discussed earlier. For the risks with higher RPN number, we plan measures to reduce the risk. The mitigation measures are also listed in the same table. Once we have reduced all risk RPN values to acceptable limit, we can finalise our design.

Circuit design

From the mitigation measures decided, we need to find the design changes that are required. Most of the recommended changes require implementation in the program logic. On the circuit design we need to provide power supply to the controller from both battery and the wind turbine. We also need to add a hooter that will get activated in the event of controller CPU stopping.

We have identified the best options for submodules and assessed various risks and made our product robust. With these inputs we can now start making the schematic circuit diagram.

The heart of the circuit is the ESP8266 based ESP12F module. This has only one ADC sensor, hence we use analogue multiplexer 75HC4051 (U3) to read the battery and wind turbine voltages. Voltages in both of these are higher than the operating range of microcontroller, hence we use voltage divider circuit comprising resistors R9, R10 and R11, R12 to reduce the voltage to the operating level.

Current flow in the power section is high. We use non-contact Hall effect sensors, U7 and U8, to read the current flowing from wind turbine and from the battery. These produce a voltage proportional to current flow, which is fed to U3.

For safety, the power controller relays will be kept separate from the main controller. We use an NTC sensor to sense any overheating in

the power module. In case of any temperature rise, the NTC resistance will drop and the voltage in the junction of R1 and R6 will rise. This is sensed by a simple voltage comparator comprising Q1, Q2, and R2 to trigger the overheating indicator LED1. The junction voltage is also fed to U3 and monitored by the CPU.

The charge controller uses an SD card to store charging and discharging data. The SD card read-write module uses SPI protocol. ESP12F has 16 GPIO pins but most of these pins are also used for internal operations. Not many pins can be used safely for control operation.

To preserve the pins, we repurpose the SPI communication pins to control analogue multiplexer. As reading analogue signals and writing values to SD card are mutually exclusive, this design is safe.

We use GPIO pins 4 and 5 to control the battery charging and brake relay operation. We use a buzzer to give warning for CPU malfunction. The buzzer is activated by a 555 timer based delay circuit.

Under normal situations the CPU will reset the delay in its operating cycle, which will be several times in every second. If the 555 is not reset in about 52 seconds, it will trigger the buzzer to give an audible warning of CPU malfunction.

The components are powered by both alternator and battery to ensure a reliable power supply. The supply voltage is around 48V. For our operations we need supply at 3.3V, 5V, and 12V. We use three DC-DC converters to get these voltages efficiently.

The addition of safety features to detect and prevent serious problems in the design will make the product more robust and improve its safety. In our next article we shall discuss how to address the effect of unknown variables and the controller logic program. **EFY**



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